

Titanic Dataset - Exploratory Data Analysis (EDA) Summary

Dataset Overview

- Total Passengers: 891
- Survived: 342 (38.4%)
- Not Survived: 549 (61.6%)

Key Insights

1. The dataset contains more male passengers (577) than female passengers (314).
2. Most passengers belonged to Third Class (491 passengers).
3. Most passengers were aged between 20 and 40 years.
4. Ticket fares are heavily right-skewed, with a few passengers paying exceptionally high fares.
5. Female passengers had significantly higher survival rates than male passengers.
6. First-class passengers were more likely to survive than second- and third-class passengers.
7. Passengers embarking from Cherbourg showed relatively better survival outcomes.
8. Higher ticket fares were associated with higher survival probabilities.
9. Passenger class and fare were the strongest factors linked to survival.
10. Age had a relatively weak influence on survival compared to gender and passenger class.

Conclusion

The analysis shows that survival on the Titanic was strongly influenced by gender, passenger class, and ticket fare. Female passengers, first-class travelers, and passengers paying higher fares had better chances of survival, while age had a comparatively smaller impact. The dataset also contains several fare and age outliers that reflect the diversity of passengers aboard the Titanic.